

SLIDING WINDOW

DSA PATTERN

From first principles to advanced counting, monotonic structures, proofs, and interview mastery

A one-stop C++ reference and learning path

Concepts | Recognition | Templates | Dry runs | Correctness | Pitfalls | 30 practice problems

How to use this guide

Read Chapters 1-5 in order if you are new. Then learn the specialized data structures in Chapters 6-8. Use the template catalog and decision tree while solving. Finish with the graded practice set and solutions.

Contents and learning roadmap

Three-stage roadmap

Basic: fixed windows and incremental state. **Intermediate:** variable windows, frequency maps, counting. **Advanced:** deques, ordered structures, transformations, proofs, and knowing when the pattern is invalid.

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1. The core idea

A sliding window reuses the work done for one contiguous range when moving to a nearby range.

1.1 From brute force to reuse

Many array and string problems ask about a **contiguous** subarray or substring. A brute-force solution chooses every left endpoint, every right endpoint, and often scans the range again. That can cost $O(n^3)$, or $O(n^2)$ if each range statistic is maintained incrementally.

A sliding window stores exactly the information needed about one active range **[left, right]**. When *right* advances, one element enters. When *left* advances, one element leaves. If each update is cheap, all useful ranges can often be processed in $O(n)$.



Active window $[1, 3]$ contains $\{1, 5, 1\}$; indices appear below.

Vocabulary

Term	Meaning
Window	The current contiguous range $[left, right]$.
Expand	Move right forward and add a new element to the state.
Shrink	Remove $a[left]$, then move left forward.
State	Sum, counts, distinct count, deque, multiset, or other summary.
Invariant	A condition guaranteed at a specific point in the loop.
Valid window	A range satisfying the problem constraint.

1.2 What makes sliding possible?

- **Contiguity:** the object is a subarray or substring, not an arbitrary subsequence.
- **Local update:** adding or removing one boundary item updates the state cheaply.
- **Useful movement rule:** we know when and why to move left or right.
- **Monotonic feasibility for variable windows:** after a violation, removing items from the left moves toward validity and discarded starts never need to return.

Important distinction

Sliding window is a specialized form of the two-pointer technique. All sliding windows use boundaries, but not every two-pointer algorithm maintains a contiguous interval. Pair-sum on a sorted array is two pointers, not usually called sliding window.

1.3 Complexity intuition: why nested loops can still be $O(n)$

Variable-window code often contains a for loop and an inner while. It is not automatically $O(n^2)$. The right pointer moves forward at most n times. The left pointer also moves forward at most n times. No pointer moves backward, so there are at most $2n$ boundary movements. This is aggregate or amortized analysis: $O(n)$ movements total.

The total complexity is $O(n * U)$, where U is the cost of one state update. An array counter or hash map is expected $O(1)$; a balanced tree or multiset makes the same traversal $O(n \log n)$.

2. Recognition and classification

Before writing code, identify the window type, validity rule, state, and answer event.

2.1 Recognition signals

- The statement says subarray, substring, contiguous segment, consecutive, or every block of size k .
- It asks for a longest, shortest, maximum, minimum, number of ranges, or whether any range satisfies a condition.
- Neighboring candidate ranges overlap heavily.
- The constraint can be repaired by moving the left boundary forward.

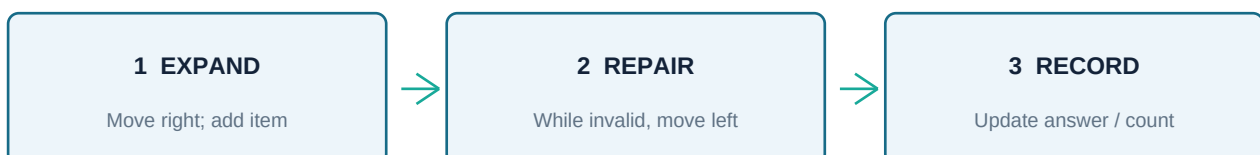
These are clues, not proof. Always test whether discarding a left endpoint is safe. Negative values, exact-equality constraints, or non-monotone properties can invalidate the ordinary variable-window method.

2.2 The main families

Family	Window size	Typical question	Core state
Fixed	Exactly k	Best/each block of length k	Sum or frequency
Variable: longest	Changes	Longest range satisfying at most...	Validity state
Variable: shortest	Changes	Shortest range reaching at least...	Validity state
Counting	Changes	How many ranges satisfy...	At-most invariant
Frequency matching	Fixed or variable	Permutation/anagram/cover	Counts + mismatch
Order statistic	Usually fixed	Maximum, minimum, median	Deque/tree/heaps

2.3 The four design questions

- 1. What does [left, right] mean?** State whether the interval is inclusive. Then length is $\text{right} - \text{left} + 1$.
- 2. What state summarizes it?** Examples: sum, number of zeros, character frequencies, distinct values, maximum-minimum, matched requirements.
- 3. What is invalid?** Write a Boolean predicate such as $\text{sum} > \text{limit}$, $\text{distinct} > k$, or $\text{max} - \text{min} > \text{limit}$.
- 4. When is the answer recorded?** After reaching size k ? After restoring validity? During shrinking? The location of this line is algorithmically important.



The core variable-window control loop

2.4 A decision checklist

Question	If yes	If no
Must the range length equal k ?	Use fixed window.	Consider variable window.
Can invalidity be repaired by removing left items?	Use expand/repair.	Seek prefix sums, DP, binary search, etc.
Are all valid suffixes ending at right countable?	Add window length.	Use another counting transform.
Need max/min under deletions?	Monotonic deque.	Simple scalar state may work.
Negative values affect a sum constraint?	Be suspicious.	Classic sum window may be valid.

3. Fixed-size windows - the foundation

Fixed windows have one deterministic movement: add the new item and remove the item that is now too old.

3.1 Maximum sum of any length-k subarray

Example: [2, 1, 5, 1, 3, 2], $k = 3$. The length-3 sums are 8, 7, 9, and 6. Instead of recomputing each sum, subtract the outgoing element and add the incoming element.

```
long long maxSumK(const vector<int>& a, int k) {
    int n = (int)a.size();
    if (k <= 0 || k > n) throw invalid_argument("invalid k");

    long long sum = 0;
    for (int i = 0; i < k; ++i) sum += a[i];
    long long answer = sum;

    for (int right = k; right < n; ++right) {
        sum += a[right]; // incoming
        sum -= a[right - k]; // outgoing
        answer = max(answer, sum);
    }
    return answer;
}
```

Invariant: immediately before updating the answer, sum equals the sum of exactly the k elements ending at right. Time $O(n)$, extra space $O(1)$. Use long long when sums may overflow int.

3.2 A universal fixed-window template

```
State state;
int left = 0;
for (int right = 0; right < n; ++right) {
    add(state, a[right]);

    if (right - left + 1 > k) {
        remove(state, a[left]);
        ++left;
    }

    if (right - left + 1 == k) {
        consume(state); // answer for this exact window
    }
}
```

This formulation is safer than manually computing outgoing indices and generalizes immediately from sums to maps, mismatch counters, and dequeues.

3.3 First negative number in every window

Maintain a deque of **indices** of negative values. Indices reveal whether an item has expired; storing only values cannot distinguish duplicates or age.

```
vector<int> firstNegative(const vector<int>& a, int k) {
    deque<int> q;
    vector<int> ans;
    for (int right = 0; right < (int)a.size(); ++right) {
        if (a[right] < 0) q.push_back(right);
        int left = right - k + 1;
        while (!q.empty() && q.front() < left) q.pop_front();
        if (left >= 0) ans.push_back(q.empty() ? 0 : a[q.front()]);
    }
    return ans;
}
```

3.4 Count distinct values in each window

```
vector<int> distinctPerWindow(const vector<int>& a, int k) {
    unordered_map<int, int> freq;
    vector<int> ans;
    for (int right = 0; right < (int)a.size(); ++right) {
        ++freq[a[right]];
        if (right >= k) {
            int out = a[right - k];
            if (--freq[out] == 0) freq.erase(out);
        }
        if (right >= k - 1) ans.push_back((int)freq.size());
    }
    return ans;
}
```

Map hygiene

Erase zero-frequency keys when map size represents the distinct count. A key with value zero still contributes to `unordered_map::size()`.

3.5 Rolling product, average, and other aggregates

Sums, counts, and XOR are easy to add and remove. Products require care: division fails with zeros, overflow is common, and modular division may not exist. An average is simply sum / k ; keep the sum in an exact integer type and convert when producing the result. Some statistics, such as a maximum, cannot be maintained with one scalar because the outgoing maximum may expose an unknown second-best value. That motivates deques and trees.

4. Variable windows - longest and shortest

The right edge explores. The left edge repairs or optimizes.

4.1 Longest subarray with sum at most S (non-negative values)

With non-negative elements, expanding can only increase the sum and shrinking can only decrease it. This monotonic behavior makes discarded starts safe.

```
int longestSumAtMost(const vector<int>& a, long long S) {
    int left = 0, answer = 0;
    long long sum = 0;
    for (int right = 0; right < (int)a.size(); ++right) {
        sum += a[right];
        while (left <= right && sum > S) {
            sum -= a[left++];
        }
        answer = max(answer, right - left + 1);
    }
    return answer;
}
```

Loop invariant after the while loop: [left, right] is valid, and every earlier left endpoint has been ruled out for this and all future right endpoints under the non-negative assumption. Therefore the current valid window is the longest valid one ending at right.

4.2 Why while, not if?

One incoming element may require removing several old elements. An if removes at most one, potentially leaving the window invalid. Use while (invalid) whenever validity is not restored by exactly one guaranteed removal.

4.3 Minimum length with sum at least S (positive values)

For a shortest-window problem, record an answer **while the window is valid** and keep shrinking to find a smaller valid range ending at the same right.

```
int minLengthAtLeast(const vector<int>& a, long long S) {
    int left = 0, answer = INT_MAX;
    long long sum = 0;
    for (int right = 0; right < (int)a.size(); ++right) {
        sum += a[right];
        while (sum >= S) {
            answer = min(answer, right - left + 1);
            sum -= a[left++];
        }
    }
    return answer == INT_MAX ? 0 : answer;
}
```

Answer placement

Longest under an upper bound: repair invalidity, then record. Shortest reaching a lower bound: record while valid, then shrink. Memorizing the code is less reliable than understanding this event.

4.4 Max consecutive ones after flipping at most k zeros

```
int longestOnes(vector<int>& a, int k) {
    int left = 0, zeros = 0, answer = 0;
    for (int right = 0; right < (int)a.size(); ++right) {
        zeros += (a[right] == 0);
        while (zeros > k) {
            zeros -= (a[left] == 0);
            ++left;
        }
        answer = max(answer, right - left + 1);
    }
    return answer;
}
```

The abstract pattern is not about zeros. It is: maintain a budget of at most k violations. The same idea solves longest substring after at most k replacements, at most k odd numbers, and at most k distinct categories.

4.5 A subtle lazy-shrink variant

Some longest-window solutions replace while with if and allow the maintained interval to remain notionally invalid, preserving only its maximum length. This can be correct for specific problems such as character replacement, but it changes the invariant and is easy to misuse. Prefer the fully valid template until you can prove the lazy version.

5. Strings and frequency-state windows

Most string windows are boundary movement plus a carefully designed frequency invariant.

5.1 Longest substring without repeating characters

Frequency-count solution

```
int lengthOfLongestSubstring(const string& s) {
    array<int, 256> freq{};
    int left = 0, answer = 0;
    for (int right = 0; right < (int)s.size(); ++right) {
        unsigned char in = s[right];
        ++freq[in];
        while (freq[in] > 1) {
            --freq[(unsigned char)s[left++]];
        }
        answer = max(answer, right - left + 1);
    }
    return answer;
}
```

Use unsigned char when indexing a 256-entry array because plain char may be signed. For lowercase English letters, 26 entries are sufficient. For Unicode text, bytes are not characters; decode code points and use a map.

Last-seen jump optimization

```
int lengthNoRepeat(const string& s) {
    array<int, 256> last;
    last.fill(-1);
    int left = 0, answer = 0;
    for (int right = 0; right < (int)s.size(); ++right) {
        unsigned char c = s[right];
        left = max(left, last[c] + 1); // never move left backward
        last[c] = right;
        answer = max(answer, right - left + 1);
    }
    return answer;
}
```

The max is essential. A character last seen before the current window must not pull left backward.

5.2 Find all anagrams of a pattern

Every answer has fixed length $|p|$. Maintain a difference count. The missing value counts how many required character instances remain, not merely how many character types remain.

```
vector<int> findAnagrams(const string& s, const string& p) {
    array<int, 256> need{};
    for (unsigned char c : p) ++need[c];
    int missing = (int)p.size();
    vector<int> ans;

    for (int right = 0, left = 0; right < (int)s.size(); ++right) {
        unsigned char in = s[right];
        if (need[in] > 0) --missing;
        --need[in];

        if (right - left + 1 > (int)p.size()) {
            unsigned char out = s[left++];
            ++need[out];
            if (need[out] > 0) ++missing;
        }
        if (missing == 0) ans.push_back(left);
    }
    return ans;
}
```

Here negative need[c] means the window contains extra copies. When an outgoing character makes need[out] positive, a required instance has become missing again.

5.3 Minimum window substring

This is a variable window with multiplicities. Expand until all required instances are covered. Then shrink while coverage remains complete, recording each improved minimum.

```
string minWindow(const string& s, const string& t) {
    if (t.empty()) return "";
    array<int, 256> need{};
    for (unsigned char c : t) ++need[c];
    int missing = (int)t.size();
    int left = 0, bestL = 0, bestLen = INT_MAX;

    for (int right = 0; right < (int)s.size(); ++right) {
        unsigned char in = s[right];
        if (need[in] > 0) --missing;
        --need[in];

        while (missing == 0) {
            if (right - left + 1 < bestLen) {
                bestLen = right - left + 1;
                bestL = left;
            }
            unsigned char out = s[left++];
            ++need[out];
            if (need[out] > 0) ++missing;
        }
    }
    return bestLen == INT_MAX ? "" : s.substr(bestL, bestLen);
}
```

5.4 Replacement windows: longest repeating character

Let maxFreq be the largest frequency of any character in the window. The number of replacements needed is windowLength - maxFreq. If this exceeds k, shrink.

```
int characterReplacement(const string& s, int k) {  
    array<int, 26> freq{};  
    int left = 0, maxFreq = 0, answer = 0;  
    for (int right = 0; right < (int)s.size(); ++right) {  
        maxFreq = max(maxFreq, ++freq[s[right] - 'A']);  
        while (right - left + 1 - maxFreq > k) {  
            --freq[s[left++]] - 'A';  
        }  
        answer = max(answer, right - left + 1);  
    }  
    return answer;  
}
```

maxFreq is intentionally not decreased during shrinking. It may be stale, but it is an upper bound representing a frequency achieved by some window of this size. This suffices for the maximum-length result. If you need the exact validity of the current window or its boundaries, recompute/maintain the true maximum instead.

6. Counting subarrays correctly

Counting windows is where a one-line combinatorial observation often replaces an extra loop.

6.1 Count ranges with an at-most constraint

After repairing the window, suppose $[left, right]$ is valid and the property is hereditary under removing items from the left. Then every suffix ending at $right$ is also valid: $[left, right]$, $[left+1, right]$, ..., $[right, right]$. There are exactly $right - left + 1$ of them.

```
long long countAtMostKDistinct(const vector<int>& a, int k) {
    if (k < 0) return 0;
    unordered_map<int, int> freq;
    int left = 0, distinct = 0;
    long long answer = 0;

    for (int right = 0; right < (int)a.size(); ++right) {
        if (freq[a[right]]++ == 0) ++distinct;
        while (distinct > k) {
            if (--freq[a[left]] == 0) {
                freq.erase(a[left]);
                --distinct;
            }
            ++left;
        }
        answer += right - left + 1;
    }
    return answer;
}
```

Counts of subarrays can reach $n(n+1)/2$, so use long long.

6.2 Exactly $k = \text{atMost}(k) - \text{atMost}(k - 1)$

The sets of ranges with at most k distinct values and at most $k-1$ distinct values differ by exactly the ranges with k distinct values.

```
long long exactlyKDistinct(const vector<int>& a, int k) {
    return countAtMostKDistinct(a, k)
        - countAtMostKDistinct(a, k - 1);
}
```

This transformation also works for exactly k odd values, exactly k zeros, exactly k violations, and other integer-valued metrics whose at-most counts are easy to compute.

6.3 Binary arrays with sum exactly goal

For non-negative/binary arrays, use $\text{count}(\text{sum} == \text{goal}) = \text{atMost}(\text{goal}) - \text{atMost}(\text{goal} - 1)$.

```
long long atMostSum(const vector<int>& a, int goal) {
    if (goal < 0) return 0;
    int left = 0, sum = 0;
    long long answer = 0;
    for (int right = 0; right < (int)a.size(); ++right) {
        sum += a[right];
        while (sum > goal) sum -= a[left++];
        answer += right - left + 1;
    }
    return answer;
}
```

6.4 Product less than k (positive values)

```
long long numSubarrayProductLessThanK(const vector<int>& a, long long k) {  
    if (k <= 1) return 0;  
    long long product = 1, answer = 0;  
    int left = 0;  
    for (int right = 0; right < (int)a.size(); ++right) {  
        product *= a[right];  
        while (product >= k) product /= a[left++];  
        answer += right - left + 1;  
    }  
    return answer;  
}
```

This assumes positive values. Zeros, negatives, and overflow change the reasoning. Even with positive integers, ensure multiplication fits the chosen type or use a problem-specific overflow guard.

6.5 A general counting theorem

Suffix-count theorem

If, after repair, left is the smallest start such that [left, right] is valid, and validity is preserved when removing elements from the left, then exactly $\text{right} - \text{left} + 1$ valid subarrays end at right. Summing this over all right endpoints counts each valid subarray once.

7. Monotonic deques for window extrema

A deque compresses all candidates that might become the maximum or minimum of a future window.

7.1 Sliding window maximum

Maintain indices in decreasing order of their values. Before inserting a new value, remove smaller or equal values from the back: the newcomer is at least as good and expires later. Remove expired indices from the front. The front is always the maximum.

```
vector<int> maxSlidingWindow(const vector<int>& a, int k) {
    deque<int> dq;
    vector<int> ans;
    for (int right = 0; right < (int)a.size(); ++right) {
        while (!dq.empty() && a[dq.back()] <= a[right]) dq.pop_back();
        dq.push_back(right);

        int left = right - k + 1;
        while (!dq.empty() && dq.front() < left) dq.pop_front();
        if (left >= 0) ans.push_back(a[dq.front()]);
    }
    return ans;
}
```

7.2 The two deque invariants

- **Index order:** indices increase from front to back. Therefore expiration happens only at the front.
- **Value order:** values decrease from front to back. Therefore the front is the maximum.
- **Dominance:** an older value no larger than a new value can never win in a future window, so it is permanently removed.

Each index is inserted once and removed at most once, giving $O(n)$ time and $O(k)$ space. For a minimum, reverse the comparison. Whether to remove equal values is a tie-policy choice; removing equals keeps the newer index, which usually expires later and is convenient.

7.3 Longest subarray with max - min $\leq$ limit

Use one decreasing deque for the maximum and one increasing deque for the minimum. Repair while their front values differ by more than the limit.

```
int longestSubarray(vector<int>& a, int limit) {
    deque<int> mx, mn;
    int left = 0, answer = 0;
    for (int right = 0; right < (int)a.size(); ++right) {
        while (!mx.empty() && a[mx.back()] <= a[right]) mx.pop_back();
        while (!mn.empty() && a[mn.back()] >= a[right]) mn.pop_back();
        mx.push_back(right);
        mn.push_back(right);

        while ((long long)a[mx.front()] - a[mn.front()] > limit) {
            if (mx.front() == left) mx.pop_front();
            if (mn.front() == left) mn.pop_front();
            ++left;
        }
        answer = max(answer, right - left + 1);
    }
    return answer;
}
```

7.4 Deque versus heap versus multiset

Structure	Update/query	Strength	Limitation
Monotonic deque	Amortized $O(1)$	Fast max/min for FIFO windows	Only extrema; careful invariant
Multiset	$O(\log k)$	Simple exact min/max and duplicates	Slower, erase one occurrence
Heap + lazy deletion	$O(\log k)$	Useful for extrema/median components	Stale entries and bookkeeping
Two multisets/heaps	$O(\log k)$	Median and quantiles	Balancing is more complex

8. Ordered windows, medians, and richer state

When add/remove is easy but the statistic is not a simple aggregate, choose a data structure that supports both boundaries.

8.1 Fixed-window median with two multisets

Maintain a lower half `lo` and upper half `hi`. Every value in `lo` is no greater than every value in `hi`. Keep `lo.size() == hi.size()` or exactly one larger. The median is the largest value in `lo`, or the average of both boundary values.

```
class MedianWindow {
    multiset<long long> lo, hi;
    void rebalance() {
        while (lo.size() > hi.size() + 1) {
            hi.insert(*prev(lo.end())); lo.erase(prev(lo.end()));
        }
        while (lo.size() < hi.size()) {
            lo.insert(*hi.begin()); hi.erase(hi.begin());
        }
    }
public:
    void add(long long x) {
        if (lo.empty() || x <= *prev(lo.end())) lo.insert(x);
        else hi.insert(x);
        rebalance();
    }
    void remove(long long x) {
        auto it = lo.find(x);
        if (it != lo.end()) lo.erase(it);
        else hi.erase(hi.find(x));
        rebalance();
    }
    double median() const {
        if (lo.size() > hi.size()) return (double)*prev(lo.end());
        return ((double)*prev(lo.end()) + (double)*hi.begin()) / 2.0;
    }
};
```

Each insertion and deletion costs $O(\log k)$, so all medians cost $O(n \log k)$. With a multiset, `erase(value)` removes every equal occurrence; use `find` and `erase` the iterator to remove exactly one.

8.2 Choosing the state representation

Need	Best first choice	Notes
Small known alphabet	<code>array<int, 26/128/256></code>	Fast, deterministic, cache friendly
Arbitrary sparse keys	<code>unordered_map</code>	Expected $O(1)$; erase zeros when needed
Ordered keys / min and max	<code>map</code> or <code>multiset</code>	$O(\log k)$, exact order
One extreme	monotonic deque	$O(n)$ total for FIFO movement
Median	two multisets/heaps	Maintain balance and ordering invariants
Bit presence for tiny universe	bitmask	Very compact; multiplicity still needs counts

8.3 Coordinate compression

If values are arbitrary but known in advance, sort unique values and map each to $[0, m)$. Then a vector can replace a hash map. This improves predictability and can be useful when many window updates are performed. Compression preserves equality and order, not original distances unless handled separately.

9. When ordinary sliding window fails

The most advanced skill is recognizing when the left pointer cannot make an irreversible safe decision.

9.1 Negative numbers break sum monotonicity

Consider longest subarray with sum at most S . With non-negative values, if the sum is too large, future expansion cannot fix it, so shrinking is safe. With negative values, a future negative number could restore validity while preserving the old left endpoint. Shrinking now may discard the optimal answer.

Counterexample

For $a = [5, -10, 5]$ and $S = 0$, shrinking immediately after seeing 5 discards index 0. But the full length-3 subarray has sum 0 and is valid. The classic variable sum window is therefore incorrect.

9.2 Exact sum with arbitrary integers: prefix frequencies

Let $\text{prefix}[j]$ be the sum of the first j elements. The sum of $[i, j-1]$ is $\text{prefix}[j] - \text{prefix}[i]$. It equals target when an earlier prefix is $\text{prefix}[j] - \text{target}$. Count earlier prefix values in a hash map.

```
long long subarraySumEqualsK(const vector<int>& a, long long k) {
    unordered_map<long long, long long> seen;
    seen[0] = 1;
    long long prefix = 0, answer = 0;
    for (int x : a) {
        prefix += x;
        if (auto it = seen.find(prefix - k); it != seen.end())
            answer += it->second;
        ++seen[prefix];
    }
    return answer;
}
```

9.3 Shortest sum at least K with arbitrary integers

Use prefix sums plus a monotonic deque of candidate prefix indices. For current prefix i , if $\text{prefix}[i] - \text{prefix}[\text{dq.front}()] \geq K$, update the answer and pop front because that start cannot produce a shorter future answer than the current one. Remove from the back while its prefix sum is no smaller than the current prefix: the newer, smaller prefix dominates it.

```
int shortestSubarray(const vector<int>& a, long long K) {
    int n = (int)a.size(), answer = n + 1;
    vector<long long> pref(n + 1);
    for (int i = 0; i < n; ++i) pref[i + 1] = pref[i] + a[i];
    deque<int> dq;
    for (int i = 0; i <= n; ++i) {
        while (!dq.empty() && pref[i] - pref[dq.front()] >= K) {
            answer = min(answer, i - dq.front());
            dq.pop_front();
        }
        while (!dq.empty() && pref[dq.back()] >= pref[i]) dq.pop_back();
        dq.push_back(i);
    }
    return answer <= n ? answer : -1;
}
```

9.4 Other warning signs

- **Subsequence rather than substring:** contiguity is absent; consider DP, greedy, or indexed positions.
- **Range must satisfy a non-hereditary property:** removing from the left may unpredictably destroy and restore validity.
- **Need arbitrary range queries:** prefix sums, sparse tables, Fenwick or segment trees may fit better.
- **Offline reordering is allowed:** Mo's algorithm moves both boundaries and answers queries, but it is not the classic one-pass sliding window.
- **Multiple independent intervals:** interval DP or sweep-line methods may be required.

9.5 Sliding window versus related techniques

Technique	Use when	Key difference
Prefix sum	Associative range aggregate or exact sums	No validity repair; subtract prefixes
Binary search + window	Feasibility is monotone in length/value	Window checks a guessed answer
Two pointers on sorted data	Pairs/triples by value order	Pointers do not summarize a contiguous original range
Monotonic stack	Next greater/smaller, spans	One directional dominance, not a moving interval
Mo's algorithm	Many offline range queries	Boundary movement is reordered, about $O((n+q)\sqrt{n})$
DP	Overlapping states without safe boundary discard	Stores more histories instead of one interval

10. Proofs, invariants, and derivation

A template becomes reliable when you can state why every pointer movement is safe.

10.1 A proof recipe for variable windows

Step 1 - Define the state. Prove add/remove operations make it exactly describe $[left, right]$.

Step 2 - State the invariant. Example: after the repair loop, the window contains at most k distinct values.

Step 3 - Prove safe shrinking. Show that every removed start cannot lead to a better candidate for the current or any relevant future right endpoint.

Step 4 - Prove answer coverage. Associate every optimal or counted range with the moment its right endpoint is processed.

Step 5 - Prove complexity. Each boundary moves forward at most n times; multiply by the data-structure update cost.

10.2 Proof sketch: longest at most k distinct

- The frequency map and distinct counter exactly describe the active interval.
- The repair loop stops only when $distinct \leq k$.
- When left advances because $distinct > k$, the discarded window is invalid. Any extension before enough categories disappear remains invalid.
- After repair, $[left, right]$ is the longest valid suffix ending at right, so updating the global maximum examines an optimal endpoint-specific candidate.
- Both pointers only advance, and each hash update is expected $O(1)$, so total expected time is $O(n)$.

10.3 Proof sketch: monotonic maximum deque

When index i is removed from the back because $a[j] \geq a[i]$ and $j > i$, index i is dominated: in every future window containing i , j is also present until i expires, and j is at least as large. Thus i can never again be the maximum. Expired indices are removed from the front. What remains is ordered by value, so the front is the correct maximum.

10.4 Derive rather than memorize

Question to ask	Example answer
What enters?	$a[right]$ increments sum/frequency/deque.
What leaves?	$a[left]$ decrements state before left advances.
What is invalid?	$distinct > k$, $zeros > k$, $max-min > limit$.
Do I repair to valid or shrink while valid?	Depends on longest vs shortest objective.
What candidates become known now?	Best ending at right, or all valid suffixes ending at right.
What assumption makes discard safe?	Non-negativity, hereditary validity, or dominance.

11. Debugging and common mistakes

Most bugs come from a broken invariant, an off-by-one boundary, or an unstated assumption.

11.1 Off-by-one rules

- Inclusive $[left, right]$ length is $right - left + 1$.
- A fixed window becomes available when $right \geq k - 1$.
- For a new fixed window ending at $right$, the expired index is $right - k$ after adding, or anything smaller than $right - k + 1$ for a deque.
- Decide whether comparisons are strict: $sum < K$ versus $sum \leq K$, $product < K$ versus $\leq K$, and $max-min \leq limit$.

11.2 Frequent implementation failures

Bug	Symptom	Fix
Use if instead of while	Window remains invalid	Shrink until invariant holds
Update answer too early	Invalid ranges included	Record only at the proven event
Forget outgoing update	State drifts from window	Remove before incrementing left
Leave zero map entries	Wrong distinct count	Erase when frequency reaches zero
Store deque values only	Cannot expire duplicates	Store indices
Move left backward	Duplicate-window bug	Use $\max(left, last+1)$
Use int for totals	Overflow	Use long long for sums/counts/products
Assume sum monotonic with negatives	Missed optimum	Use prefix-based method
<code>multiset.erase(value)</code>	All duplicates erased	Erase one iterator
$k=0$ / $k>n$ unhandled	Undefined semantics	Validate constraints explicitly

11.3 Trace table method

For each right endpoint, write: incoming item, state after add, removals, final left, validity, and answer change. This exposes almost every pointer bug.

right	in	state after add	left removals	final $[l,r]$	answer
0	2	sum=2	-	$[0,0]$	1
1	1	sum=3	-	$[0,1]$	2
2	5	sum=8	remove 2,1	$[2,2]$	2
3	1	sum=6	-	$[2,3]$	2
4	1	sum=7	-	$[2,4]$	3

Example assumes longest sum ≤ 7 with non-negative numbers.

11.4 Edge-case test suite

- Empty input; one element; all equal; all distinct; all zeros.
- $k = 0, 1, n, n+1$; target negative/zero; no valid window; whole array valid.
- Duplicate values at both boundaries; optimum at beginning or end.
- Large values causing overflow; characters outside expected alphabet.
- For monotonic deque: strictly increasing, strictly decreasing, and all equal.

12. C++ template catalog

Use these as skeletons. Replace State, validity, and answer logic only after writing the invariant.

12.1 Fixed-size template

```
int left = 0;
State state;
for (int right = 0; right < n; ++right) {
    add(state, a[right]);
    if (right - left + 1 > k) remove(state, a[left++]);
    if (right - left + 1 == k) record(state, left, right);
}
```

12.2 Longest valid template

```
int left = 0, answer = 0;
State state;
for (int right = 0; right < n; ++right) {
    add(state, a[right]);
    while (invalid(state)) remove(state, a[left++]);
    answer = max(answer, right - left + 1);
}
```

12.3 Shortest valid template

```
int left = 0, answer = INF;
State state;
for (int right = 0; right < n; ++right) {
    add(state, a[right]);
    while (valid(state)) {
        answer = min(answer, right - left + 1);
        remove(state, a[left++]);
    }
}
```

12.4 Count all hereditary-valid windows

```
int left = 0;
long long answer = 0;
State state;
for (int right = 0; right < n; ++right) {
    add(state, a[right]);
    while (invalid(state)) remove(state, a[left++]);
    answer += right - left + 1;
}
```

12.5 Exactly-k transform

```
long long exactlyK(const Input& x, int k) {
    return atMost(x, k) - atMost(x, k - 1);
}
```

12.6 Monotonic deque template

```
deque<int> dq;
for (int right = 0; right < n; ++right) {
    while (!dq.empty() && dominated(dq.back(), right)) dq.pop_back();
    dq.push_back(right);
    int left = right - k + 1;
    while (!dq.empty() && dq.front() < left) dq.pop_front();
    if (left >= 0) record(a[dq.front()]);
}
```

12.7 Binary search on answer plus fixed window

Sometimes the question asks for the maximum feasible length and feasibility is monotone by length. Binary-search the length, and test each length with a fixed window. This often costs $O(n \log n)$. Before using it, check whether a direct variable window can already solve the problem in $O(n)$.

13. Advanced patterns and extensions

Sliding window is a framework: boundary movement can be combined with transformations, dequeues, hashing, and binary search.

13.1 Complement window: take from both ends

If you must take exactly k items from the two ends of an array, choosing those items is equivalent to leaving one contiguous middle window of length $n-k$. Maximize taken sum by minimizing the sum of that fixed middle window. This complement transformation turns an awkward two-ended choice into a window.

13.2 Circular arrays

A circular window of length at most n can be processed over indices $0..2n-1$ using $a[i \% n]$, while preventing length from exceeding n . For fixed $k \leq n$, scan only enough doubled positions to enumerate n starting points. Avoid physically duplicating a large array unless clarity matters more than memory.

13.3 Multiple constraints

Combine state components: sum plus distinct count, max/min dequeues plus a budget, or several character requirements. The invalid predicate becomes an OR of violations. Ordinary variable windows remain valid only if shrinking moves every violated constraint monotonically toward validity.

13.4 Two windows in one pass for exactly- k counting

Instead of calling `atMost` twice, maintain two left boundaries: one for at most k and one for at most $k-1$. At each right, the number of exactly- k ranges ending there is `leftKMinus1 - leftK`. This can reduce constant factors but is harder to reason about; the two-pass identity is usually cleaner.

13.5 Frequency of the maximum element

Problems may ask for subarrays where the global maximum appears at least k times. Once the active window contains k copies, every extension of its left boundary in the opposite direction may be counted. Carefully identify whether you are counting valid prefixes or valid suffixes; not every counting problem uses `right-left+1`. Derive the combinatorics from the endpoint fixed by the loop.

13.6 Windows over streams

A fixed window naturally supports streaming: keep only the last k items or the state needed to expire them. Time-based windows require timestamps and expire all events older than the cutoff. Out-of-order events need watermark/buffering semantics and go beyond the simple array model.

13.7 2D sliding windows

For matrix rectangle sums, a 2D prefix sum usually gives $O(1)$ per rectangle. For row-wise/column-wise extrema, apply monotonic dequeues in two passes. A k by k maximum filter can compute row maxima of width k , then column maxima of height k , in $O(\text{rows} * \text{cols})$.

13.8 Smallest range covering k sorted lists

Merge all values with their list ID, sort them, then find the smallest window containing every list ID. This is minimum-window substring generalized from characters to categories. A heap-based k-way merge is another approach.

13.9 Offline queries and Mo's algorithm

Mo's algorithm orders arbitrary range queries so consecutive queries require few add/remove operations. Unlike the classic pattern, both boundaries may move in both directions, and correctness does not rely on monotone validity. It is useful when updates are cheap but no fast per-query data structure exists.

14. Worked pattern map

Different stories often reduce to the same invariant.

Problem phrase	Invariant / transformation	Answer event
Maximum average length k	Fixed size; rolling sum	When size == k
Longest no repeats	Every frequency ≤ 1	After repair
Longest at most k distinct	distinct $\leq k$	After repair
Shortest covering pattern	missing == 0	While valid
Permutation in string	Fixed size + matching frequencies	When size == p
Count exactly k odds	atMost(k) - atMost(k-1)	Add valid suffix count
Max of every length k	Decreasing index deque	When size == k
Longest max-min $\leq L$	Two monotonic deques	After repair
Exact sum, arbitrary ints	Prefix frequency map	At each prefix
Shortest sum $\geq K$, arbitrary ints	Prefix + monotonic deque	Pop valid fronts
Take k cards from ends	Minimize complement window n-k	When size == n-k
Cover k sorted lists	Window over merged (value, list)	While all lists present

14.1 Interview communication script

- State the brute force and its complexity.
- Point out the overlap between neighboring contiguous ranges.
- Define [left, right], the maintained state, and the validity predicate.
- Explain exactly when right expands and when left shrinks.
- State the invariant before coding.
- After coding, justify correctness, $O(n)$ amortized movement, update cost, and space.
- Name assumptions explicitly: non-negative values, alphabet, k bounds, overflow type.

15. Graded practice set

Solve in order. For each problem, first write the family, state, invalid predicate, answer event, and complexity without code.

15.1 Basic

#	Problem	Main lesson
1	Maximum sum subarray of size k	Rolling fixed state
2	Average of every subarray of size k	Same state, different output
3	First negative in every window	Queue indices and expiration
4	Count distinct in each window	Frequency map lifecycle
5	Maximum number of vowels in length k substring	Fixed violation counter
6	Find all anagrams	Fixed frequency matching

15.2 Intermediate

#	Problem	Main lesson
7	Longest substring without repeats	Repair duplicates
8	Longest subarray with at most k zeros	Budget invariant
9	Fruit into baskets / at most two types	At most k distinct
10	Minimum size sum $\geq S$, positive array	Shrink while valid
11	Minimum window substring	Multiplicity and matched count
12	Count product $< k$, positive array	Count valid suffixes
13	Count exactly k distinct	Difference of at-most counts
14	Binary subarrays with sum goal	Exactly via at-most transform
15	Longest repeating character replacement	Dominant frequency

15.3 Advanced

#	Problem	Main lesson
16	Sliding window maximum	Monotonic deque
17	Longest range with max-min $\leq$ limit	Two deques
18	Sliding window median	Balanced ordered halves
19	Shortest subarray sum $\geq K$ with negatives	Prefix monotonic deque
20	Smallest range covering k lists	Merged categorical window
21	Subarrays with exactly k odd numbers	Exactly transform
22	Maximum points from taking k ends	Complement window
23	Substring concatenation of all words	Multiple offset windows
24	Minimum operations to reduce x to zero	Longest complement sum
25	Count complete subarrays	At-most or ending-point count

15.4 Expert extensions

#	Problem	Direction
26	2D k x k maximum filter	Two-pass monotonic deques
27	Time-based event-rate window	Timestamp deque and expiry
28	Median absolute deviation in fixed windows	Ordered partitions + sums
29	Many offline distinct-range queries	Mo's algorithm
30	Design a generic reusable window aggregator	Interfaces, invariants, tests

16. Hints and compact solutions

Use these only after attempting the classification and invariant yourself.

16.1 Problems 1-6

1-2. Initialize the first k sum or use the universal fixed template. For averages, divide only when emitting. **3.** Store negative indices; expire indices smaller than left. **4.** Increment incoming, decrement outgoing, erase zero. **5.** Track a vowel count. **6.** Maintain a difference array or missing-instance counter over exact-size windows.

16.2 Problems 7-15

7. Frequency repair or last-seen jump. **8.** Invalid when zeros > k. **9.** Invalid when map size > 2. **10.** Record while sum >= S, then shrink; positivity is required. **11.** Count missing required instances. **12.** Repair product >= k and add window length. **13.** atMost(k) - atMost(k-1). **14.** Same transform because binary values are non-negative. **15.** replacements = length - largest frequency.

16.3 Problems 16-25

16. Decreasing deque of indices. **17.** Decreasing max deque plus increasing min deque. **18.** Two balanced multisets or heaps with lazy deletion. **19.** Prefix sums in an increasing deque. **20.** Sort merged (value,listID) pairs, then cover all IDs. **21.** Treat odd as 1, even as 0; use at-most difference. **22.** total minus minimum length n-k middle sum. **23.** Run one word-aligned window per offset modulo word length. **24.** Find longest subarray with sum total-x; positivity enables a normal window. **25.** Count windows containing all globally distinct values by shrinking while complete and counting possible starts.

16.4 Problems 26-30

26. Compute width-k row maxima, then height-k column maxima. **27.** Expire timestamps earlier than now-duration. Define inclusive cutoff precisely. **28.** Median minimizes absolute deviation; maintain two ordered halves plus sums of each half. **29.** Sort queries by blocks and update frequency state as boundaries move. **30.** Expose add/remove/query, document valid movement directions, and property-test against brute force.

16.5 A reusable brute-force oracle

For small random arrays, compare your optimized result with an obviously correct $O(n^2)$ implementation. This is one of the fastest ways to validate window invariants and boundary conditions.

```
// Example oracle: longest subarray with sum <= S.
int brute(const vector<int>& a, long long S) {
    int ans = 0;
    for (int l = 0; l < (int)a.size(); ++l) {
        long long sum = 0;
        for (int r = l; r < (int)a.size(); ++r) {
            sum += a[r];
            if (sum <= S) ans = max(ans, r - l + 1);
        }
    }
    return ans;
}
```

17. Final cheat sheet

The whole pattern, compressed into one reference page.

Recognition

Contiguous range + overlapping candidates + cheap add/remove + safe monotone boundary movement. If any part is missing, reconsider the technique.

Core formulas

Idea	Formula / rule
Inclusive length	$\text{right} - \text{left} + 1$
Valid suffixes ending at right	$\text{right} - \text{left} + 1$
Exactly k	$\text{atMost}(k) - \text{atMost}(k - 1)$
Fixed outgoing index after adding right	$\text{right} - k$
Replacement cost	window length - maximum frequency
Subarray sum by prefixes	$\text{prefix}[\text{r} + 1] - \text{prefix}[\text{l}]$
All subarrays count	$n(n + 1) / 2$

Answer-event map

Goal	When to update
Best exact length k	After the active size becomes k
Longest under an upper bound	After repairing invalidity
Shortest reaching a lower bound	Inside the while-valid shrink loop
Count hereditary-valid ranges	After repair: add current window length
Minimum cover	Inside the complete-cover shrink loop

Complexity

Classic window: $O(n)$ boundary movements. With state update U: $O(nU)$. Hash/array state usually $O(n)$ total; ordered state $O(n \log k)$; monotonic deque $O(n)$ amortized. Space ranges from $O(1)$ to $O(k)$ or $O(\text{alphabet})$.

The one sentence to remember

Expand to explore, shrink to restore or optimize, maintain only what the active interval needs, and never discard a boundary unless you can prove it cannot matter later.

Mastery test

You have mastered sliding window when you can: recognize it without keywords; derive rather than memorize the loop; state the invariant; explain answer placement; prove pointer movements; identify negative-number failures; choose the right state structure; and reduce a new story to a known window family.

Seven-day mastery plan

Day	Focus	Deliverable
1	Core idea + fixed windows	Solve practice 1-5 and trace two by hand
2	Variable longest/shortest	Solve 7-10; write the invariant first
3	String frequency state	Solve 6, 11, and 15 without copying templates
4	Counting transformations	Solve 12-14 and explain right-left+1
5	Monotonic dequeues	Solve 16-17; prove every deque pop is safe
6	Failure cases + advanced state	Solve 18-20 and compare alternatives
7	Mixed timed revision	Pick six unseen problems; classify before coding

Before submitting any window solution

- Check empty input, k bounds, strict versus non-strict comparisons, and overflow.
- Verify state exactly matches [left, right] after every add and remove.
- Confirm the answer line occurs only when its candidate is mathematically known.
- State the assumption that makes moving left irreversible and safe.
- Test against a brute-force oracle on small random and adversarial inputs.